UNIVERSITY OF NUEVA CACERES
SCHOOL OF COMPUTER AND INFORMATION SCIENCES
INTELLIGENT SYSTEMS/ INFORMATION SYSTEMS

SYSTEM PROPOSAL

MIDTERM PROJECT

A. PROJECT INFORMATION

Proposed System Title	Real Property Assessment Tracking and Inter-Office Synchronization System
Researchers / Developers	Christian Louise O. Asido, John Kenneth P. Calubad, Rizalina A. Ortega
Program / Course / Section	Bachelor of Science in Information Technology (BSIT)
Client / Beneficiary / Organization	Municipal Assessor's Office and Municipal Treasurer's Office — Canaman, Magarao, San Fernando, and Pasacao, Camarines Sur (with Provincial Assessor's Office approval review in Pili)
Adviser / Instructor	Jay Leandro Borela
Date Submitted	September 2026

B. PROPOSAL OUTLINE

1. PROJECT TITLE

Provide a concise and specific title that identifies the proposed system, its main purpose, and/or its target users.

Proposed title: Needs Assessment for a Real Property Assessment Tracking and Inter-Office Synchronization System: Resolving Disconnected Records and Workflow Delays in Municipal Assessor and Treasurer Offices in Pasacao

2. PROJECT BACKGROUND / CONTEXT

Explain the current situation or environment where the system will be used. Describe the organization, users, existing process, and relevant context.

Key points to discuss:
• Current situation or process
• Organization/client context
• Target users
• Relevant observations or evidence

Local government units in the Philippines collect real property taxes as a primary revenue source. This function requires close administrative coordination among municipal and provincial offices. The Municipal Assessor's Office appraises property values. The Municipal Treasurer's Office collects tax payments. Under Republic Act 7160, the Provincial Assessor's Office in Pili holds final approval authority over property assessments and transfers of ownership. In practice, paper-based approval steps cause long operational delays across these offices.

The target environment includes four municipalities in Camarines Sur: Canaman, Magarao, San Fernando, and Pasacao. These fourth-class and lower-income municipalities operate with limited computer hardware and small technical teams. Some paper archives date back to 1949. Pasacao serves as the primary benchmark site. Pasacao operates a local Electronic Real Property Tax (eRPT) installation for internal assessor records. However, a Transfer of Assessment in Pasacao requires more than seven days. In Canaman and Magarao, personnel manage all records on paper. An assessment of a new building or a transfer of title requires 7 to 14 days because staff must physically transport paper forms to the Provincial Capitol in Pili for final approval.

Target users:

- **Municipal Assessor's Office personnel:** Staff who encode property assessments and assemble title documents.
- **Municipal Treasurer's Office personnel:** Staff who review tax payments and issue financial clearances.
- **Provincial Assessor's Office personnel:** Officers in Pili who examine ownership documents and grant final statutory approvals.
- **Municipal administrators and supervisors:** Officers who monitor transaction queues across offices.

The research team observed processing workflows and interviewed municipal staff during the needs assessment phase. Neither office has real-time visibility into transaction statuses. Documents stop in manual queues. Staff spend hours verifying whether a taxpayer paid property tax. These conditions match the findings of Chinilla and Lagura (2025). Their study confirms that paper records cause administrative delays in municipal offices. Sunirah et al. (2026) also show that isolated tax applications do not reduce workflow backlogs.

3. STATEMENT OF THE PROBLEM

Identify the specific problems, limitations, inefficiencies, or unmet needs that the proposed system will address.

Main Problem: The lack of a shared digital platform forces municipal and provincial offices to use manual procedures. Consequently, assessment processing requires 7 to 14 days, and staff must manually cross-check records and physically transport documents to Pili for final approval.

Specific Problems:

1. **Disconnected Departmental Databases:** The Assessor's Office and the Treasurer's Office keep records in separate systems or paper ledgers. Staff must carry physical documents or call other offices to reconcile record discrepancies.
2. **Document Processing Latency:** A Transfer of Assessment in Pasacao requires more than seven days because staff transfer paper documents manually. In Canaman, title transfers and building assessments take 7 to 14 days because staff transport paper forms to Pili for provincial approval.
3. **Manual Provincial Approval Process:** The Provincial Assessor's Office in Pili holds statutory final approval for property assessments and ownership title changes. Currently, municipal offices rely on physical couriers to deliver paper packets to Pili, which causes severe transit delays.
4. **Lack of Document Tracking Visibility:** Offices cannot track the location of an assessment document in the processing queue. Frontline staff must make phone calls or search file cabinets to answer taxpayer questions.
5. **Legacy Paper Records:** Some municipalities store paper records that date back to 1949. These physical binders deteriorate over time. They slow down verification tasks and make record updates difficult.
6. **Manual Cross-Checking:** Treasurer personnel manually compare tax payment entries against assessor rolls. This manual process takes many work hours and increases error rates during tax collection periods.

4. PROPOSED SOLUTION

Describe the proposed system and explain how it will address the identified problems. Give a clear overview of what the system will do.

System overview: The proposed system is a web-based Real Property Assessment Tracking and Inter-Office Synchronization System. The system uses a three-tier web architecture. It provides a shared web application for municipal offices and an endorsement portal for the Provincial Assessor's Office in Pili.

The platform contains three functional tiers:

- **Presentation Tier:** Provides role-based browser screens. These include assessor evaluation screens, treasurer validation views, a provincial approval portal, and a document tracking monitor.
- **Application Tier:** Manages business logic. This tier routes assessment documents automatically, controls user permissions with Role-Based Access Control (RBAC), links valuations to tax records, and manages digital provincial endorsements.
- **Data Tier:** Stores all records in a centralized relational database. This tier maintains property assessment data, ownership title attachments, transaction logs, and audit records.

The system will operate in parallel with existing office procedures during pilot testing. This parallel run protects data integrity and validates software stability. In compliance with Republic Act 10173 (Data Privacy Act of 2012), developers will use masked dummy data during system development and testing.

5. PROJECT OBJECTIVES

State what the project intends to accomplish. Objectives should be specific, measurable where possible, and directly related to the identified problems.

General Objective: Design, develop, and evaluate a web-based Real Property Assessment Tracking and Inter-Office Synchronization System that connects municipal assessor and treasurer offices, automates document routing to the provincial office, and reduces processing delays.

Specific Objectives:

1. Conduct a needs assessment across Pasacao, Canaman, Magarao, and San Fernando to measure baseline processing times and record system requirements.
2. Design and develop a three-tier web platform with assessor screens, treasurer validation views, a provincial digital approval module, and a centralized database.
3. Implement an inter-office web portal that provides real-time document tracking, eliminates physical courier trips to Pili, and removes manual cross-checking between offices.
4. Evaluate the software under ISO/IEC 25010 standards for Functional Suitability, Performance Efficiency, Usability, Security, and Reliability, and measure reductions in processing time.

6. SCOPE AND DELIMITATIONS

Define what the proposed system will cover and what it will not cover. Clearly state the boundaries of the project.

Scope:

- The system covers property assessment tracking, document routing, payment synchronization, provincial digital approval, and administrative monitoring.
- The study covers four municipalities in Camarines Sur: Pasacao, Canaman, Magarao, and San Fernando, with provincial endorsement review in Pili.
- Baseline data collection involves time-motion measurements of current assessment, tax collection, and provincial approval workflows.
- The system supports property assessment creation, Transfer of Assessment tracking, ownership title verification, provincial approval, and payment status updates.
- The development process uses an Agile method with two-week sprint cycles.
- The project evaluation runs from July 2026 to December 2026.
- The software runs on standard computers connected to a municipal network and allows web access for provincial approvers.

Delimitations / Exclusions:

- The system does not process online payments or connect to external financial payment gateways.
- The system does not rebuild the internal enterprise database of the Provincial Capitol in Pili; it provides a web-based review and approval portal.
- The system excludes business permits, civil registry records, and general municipal accounting.
- Evaluation results reflect conditions in lower-class municipalities and may not apply to large urban cities.
- The project does not use live citizen property records during development and testing, in compliance with Republic Act 10173.

7. TARGET USERS / STAKEHOLDERS

Identify the people or groups who will use, manage, benefit from, or be affected by the system.

USER / STAKEHOLDER	ROLE / NEED	EXPECTED BENEFIT
Assessor's Office Personnel	Create assessment records, attach title documents, route files, and monitor transaction stages	Eliminates paper handoffs and enables real-time document tracking

USER / STAKEHOLDER	ROLE / NEED	EXPECTED BENEFIT
Treasurer's Office Personnel	Review assessment records, validate tax payments, and issue clearances	Removes manual cross-checking and receives assessment updates automatically
Provincial Assessor Personnel (Pili)	Review submitted assessment packets, inspect ownership titles, and grant digital approvals	Eliminates physical courier delays and provides digital endorsement tools
Municipal Administrators / Supervisors	Monitor inter-office queues, view audit logs, and inspect performance	Provides clear visibility into assessment queues and transaction histories
Property Owners and Local Residents	Submit assessment requests, transfer property titles, and inquire about status	Shortens waiting periods from 14 days to a few days and reduces office visits
Local Government Units of Camarines Sur	Meet digital governance goals under Republic Act 12001 and national masterplans	Establishes an auditable digital workflow between municipal and provincial offices
Future IT Researchers	Study public sector data synchronization and multi-tier government workflows	Provides measured baseline data and a reference system architecture

8. SIGNIFICANCE / BENEFITS OF THE PROJECT

Explain the expected benefits of the proposed system to the client, users, organization, community, or other stakeholders.

Expected benefits:

- **Assessor's Office Personnel:** Staff do not need to carry physical tax forms or search old paper archives. Personnel encode records once and track each document automatically through the routing pipeline.
- **Treasurer's Office Personnel:** Real-time record synchronization removes the need to cross-check ledgers manually. The system validates payment records automatically.
- **Provincial Assessor's Office Personnel:** Officers review ownership documents and grant approvals digitally. This process eliminates courier trips and lost paper files.
- **Property Owners and Local Residents:** The system reduces processing delays for transactions like a Transfer of Assessment. Citizens make fewer office visits to track document status.
- **Local Government Units:** The software provides an auditable digital workflow. This workflow supports national e-governance policies under Republic Act 12001 and the Digital Government Masterplan 2023–2028.
- **Future IT Researchers:** The study provides measured performance data and an architectural model for inter-office synchronization in rural government offices.
- **Academic Institutions:** The project supports the Institutional Research Agenda of the University of Nueva Caceres regarding digital public service transformation.

9. EXISTING SYSTEM / CURRENT PROCESS

Describe how the task or service is currently performed. Identify existing tools, procedures, manual processes, or software being used.

Current process description: In Pasacao, the Municipal Assessor's Office uses an internal eRPT software tool. This tool does not connect to the Treasurer's Office. When a citizen requests a Transfer of Assessment, assessor staff print paper forms. Staff physically deliver these forms to the Treasurer's Office. Treasurer personnel verify payment history against a separate ledger. The documents then return to the Assessor's Office for final encoding. This manual exchange takes more than seven days.

In Canaman, Magarao, and San Fernando, municipal offices do not use digital tax applications. Staff process all property records and tax clearances on paper. To assess a new building or transfer title of ownership, personnel encode paper forms and send them to the Provincial Capitol in Pili. This approval process requires 7 to 14 days because physical couriers transport paper folders. Staff in both municipal offices cross-check assessment rolls and collection ledgers by hand.

None of the four municipalities have a shared document tracking dashboard or automated inter-office routing to the provincial level. Staff rely on telephone calls and office visits to check document status. Documents sometimes stop in transit without notice.

10. PROPOSED SYSTEM FEATURES AND FUNCTIONS

List the major features and functions. Explain what each feature does and which user or problem it addresses.

FEATURE / FUNCTION	DESCRIPTION	USER
Assessment Encoding Module	Lets assessor staff enter property data and upload ownership title attachments into the shared database	Assessor Staff
Automated Document Routing Engine	Transfers assessment documents through defined review steps between municipal and provincial offices automatically	Assessor Staff, Treasurer Staff, Provincial Staff
Real-Time Document Tracking Monitor	Displays current transaction status and assigned personnel on a shared dashboard	All Users, Administrators
Tax Payment Synchronization Module	Matches treasurer payment entries with assessor property records automatically	Treasurer Staff
Treasurer Validation View	Lets treasurer staff review incoming assessment files and record clearances digitally	Treasurer Staff
Provincial Review and Digital Approval Module	Enables provincial assessor officers in Pili to review ownership titles, inspect valuation sheets, and grant final digital approval	Provincial Assessor Personnel
Role-Based Access Control (RBAC)	Limits system access based on user role to protect citizen records under Republic Act 10173	System Administrator
Audit Trail and Transaction Log	Records all document changes and approval actions with timestamps and user names for accountability	Administrators, Supervisors
Administrative Workflow Panel	Lets supervisors monitor workflow queues, manage accounts, and generate activity reports	Administrators

11. SYSTEM REQUIREMENTS

Identify the expected functional and non-functional requirements.

Functional requirements: What the system must do.

1. The system must allow assessor staff to create property assessment records and attach ownership title documents through a web interface.
2. The system must route submitted assessment records to the Treasurer's Office queue automatically.
3. The system must provide a treasurer interface to review assessment documents and record clearances.
4. The system must route cleared assessment packets to the Provincial Assessor's Office queue in Pili for final review.
5. The system must allow provincial assessor officers to inspect ownership documents and record final digital approvals.
6. The system must display a real-time dashboard that shows transaction status to all participating offices.
7. The system must match property assessment files with tax payment entries and flag mismatches.
8. The system must restrict functions and data views based on assigned user roles.
9. The system must record an audit log with timestamps and user identities for all document actions.
10. The system must support a parallel run mode that works alongside current manual office workflows.

Non-functional requirements: Performance, security, usability, reliability, compatibility, etc.

1. **Performance Efficiency:** The system must load dashboard views and routing updates within three seconds on local and remote network connections.
2. **Usability:** Staff must be able to complete routine encoding, validation, and approval tasks within five user actions.
3. **Security:** The system must encrypt stored property records, use HTTPS for data transfer, and enforce session timeouts.
4. **Reliability:** The system must use atomic database transactions to prevent incomplete data updates.
5. **Compatibility:** The web application must function on Google Chrome, Mozilla Firefox, and Microsoft Edge without extra plugins.
6. **Functional Suitability:** The software must perform all defined routing, matching, and approval operations without logic errors.

12. TECHNOLOGY / PLATFORM

Identify the proposed hardware, software, programming languages, frameworks, database, platforms, APIs, or other technologies.

Proposed technologies:

- **Hardware:** Standard desktop computers in municipal offices and the Provincial Capitol, connected via local and wide area networks.
- **Software / Tools:** Standard web browsers (Google Chrome, Mozilla Firefox, Microsoft Edge) and a secure web application server.
- **Programming / Frameworks:** PHP (Laravel or CodeIgniter framework) for backend logic, with HTML5, CSS3, and JavaScript for user interfaces.
- **Database:** MySQL or MariaDB relational database for property records, payment logs, and audit data.
- **Platform:** Web application deployed on a municipal server with secure remote portal access for provincial approvers.

13. PROPOSED SYSTEM ARCHITECTURE / CONCEPTUAL DESIGN

Present a high-level model of how users, interfaces, application components, databases, external services, and other components interact.

The system uses a three-tier web architecture. This design separates the user interface, business logic, and database operations into three distinct layers.

TIER	LAYER	COMPONENTS
Tier 1 — Presentation	Frontend (Browser)	Assessor Screen, Treasurer View, Provincial Approval Portal, Tracking Dashboard, Admin Panel
Tier 2 — Application	Backend (Business Logic)	Document Routing Engine, Provincial Approval Logic, RBAC Module, Payment Synchronization, Audit Trail Manager
Tier 3 — Data	Database	Centralized Relational Database (MySQL or MariaDB) storing property records, title attachments, approval logs, and payment entries

Data Flow Summary:

1. Assessor staff enters property data and attaches ownership title documents in the browser interface.
2. The server application validates the data and writes the record to the central database.
3. The routing engine sends the document to the Treasurer's Office queue.
4. Treasurer staff reviews the file and marks the tax payment status.
5. The routing engine transfers cleared packets to the Provincial Assessor's Office queue in Pili.
6. Provincial personnel review ownership documents and grant final digital approval.
7. The application updates the database and refreshes the tracking dashboard for all offices.

Required Diagrams:

a. System Architecture Diagram:

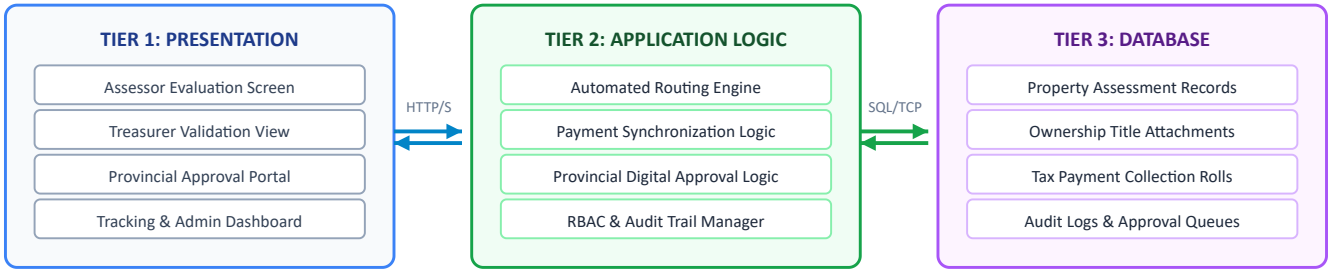


Figure 1. Three-Tier System Architecture Diagram (with Provincial Approval Tier)

b. Systems Flowchart:

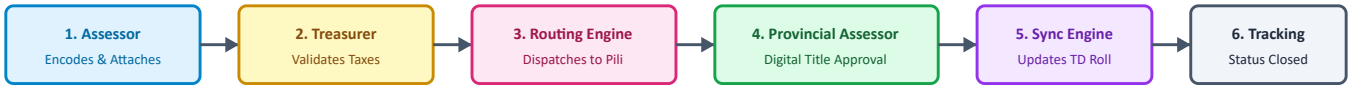


Figure 2. End-to-End Inter-Office & Provincial Assessment Systems Flowchart

14. DEVELOPMENT METHODOLOGY

Identify and briefly explain the development methodology or framework to be used (e.g., Agile, Scrum, Waterfall, Prototype, RAD). Explain why it is appropriate for the project.

Methodology: Agile Development Model (two-week sprint cycles)

Reason for selection: The Agile model allows the team to adapt software features based on ongoing feedback from municipal personnel and provincial approvers. The project begins with a field needs assessment that uncovers specific workflow requirements across offices. Two-week sprints let developers deliver testable software increments regularly. Municipal and provincial staff can test each increment, such as the routing engine, treasurer validation view, and provincial approval module, before the team builds subsequent features. This approach reduces design errors before pilot deployment. It also supports the mixed-methods research design through regular evaluation intervals.

15. PROJECT DEVELOPMENT PLAN / TIMELINE

Provide the major project activities, expected outputs, responsible members, and target schedule.

ACTIVITY	EXPECTED OUTPUT	RESPONSIBLE MEMBER	TARGET DATE
1. Needs Assessment & Field Study	Baseline latency metrics and documented workflow bottlenecks across municipal and provincial offices	All Members	July 2026
2. Requirements Analysis & Design	Requirements specification document and multi-office system design diagrams	All Members	August 2026
3. Sprint 1: Database Setup & Backend	Relational database schema with title attachments and user login system with RBAC	Calubad, Asido	Aug – Sept 2026
4. Sprint 2: Assessment Encoding & Routing	Assessment input form with document uploads and automated inter-office routing engine	Calubad, Asido	September 2026

ACTIVITY	EXPECTED OUTPUT	RESPONSIBLE MEMBER	TARGET DATE
5. Sprint 3: Treasurer Validation & Sync	Treasurer validation interface and automated payment record matching	Asido, Calubad	October 2026
6. Sprint 4: Provincial Approval Portal	Provincial digital review screen for title verification and endorsement logging	Calubad, Ortega	October 2026
7. Sprint 5: Tracking Dashboard & Audit	Real-time document tracking screen and user activity audit log across all offices	Calubad, Ortega	October – Nov 2026
8. Integration & Internal Testing	Integrated software increment and internal test report using dummy data	All Members	November 2026
9. Pilot Deployment & Parallel Run	Pilot installation in offices and staff training completion report	All Members	November 2026
10. ISO/IEC 25010 Evaluation & Defense	Evaluation findings report, completed BSIT capstone manuscript, and defense demo	All Members	December 2026

REFERENCES

Anti-Red Tape Authority. 2022. *Memorandum Circular No. 2022-02: Compliance on Automatic Approval, Zero-Contact Policy, and Zero Backlog Program.* Republic of the Philippines.

Norberto Arrocena Chinilla and Glenne Berja Lagura. 2025. Enhanced tax revenue assessment and collection system through the lens of the municipal treasurers and assessors. *International Journal for Multidisciplinary Research* 7, 4 (2025), 1–42. DOI: https://doi.org/10.36948/ijfmr.2025.v07i04.51726

Department of Information and Communications Technology. 2023. *Digital Government Masterplan (DGMP) 2023–2028.* Republic of the Philippines.

International Organization for Standardization. 2025. *ISO 19152-4:2025: Geographic Information — Land Administration Domain Model (LADM) — Part 4: Valuation Information.* ISO.

Raúl Félix Junquera-Varela and Cristian Óliver Lucas-Mas (Eds.). 2024. *Revenue Administration Handbook.* World Bank. DOI: https://doi.org/10.1596/978-1-4648-2053-3

Xaver Schenker and Wilson Prichard. 2024. *Key Elements of IT System Design for Property Taxation.* LoGRI Guidance Note No. 04. Local Government Revenue Initiative, University of Toronto.

Nuroh Sunirah, Kridawati Sadhana, Roos Widjajani, and Ratih Nindita Kirana. 2026. Evaluating the effectiveness of a land and building tax information system in improving property tax collection: Evidence from Madiun City, Indonesia. *International Journal of Management and Economics Invention* 12, 7 (2026), 5844–5859. DOI: https://doi.org/10.47191/ijmei/v12i7.09

Republic of the Philippines. 1991. *Republic Act No. 7160: Local Government Code of 1991.* Manila.

Republic of the Philippines. 2012. *Republic Act No. 10173: Data Privacy Act of 2012.* Manila.

Republic of the Philippines. 2024. *Republic Act No. 12001: Real Property Valuation and Assessment Reform Act.* Manila.